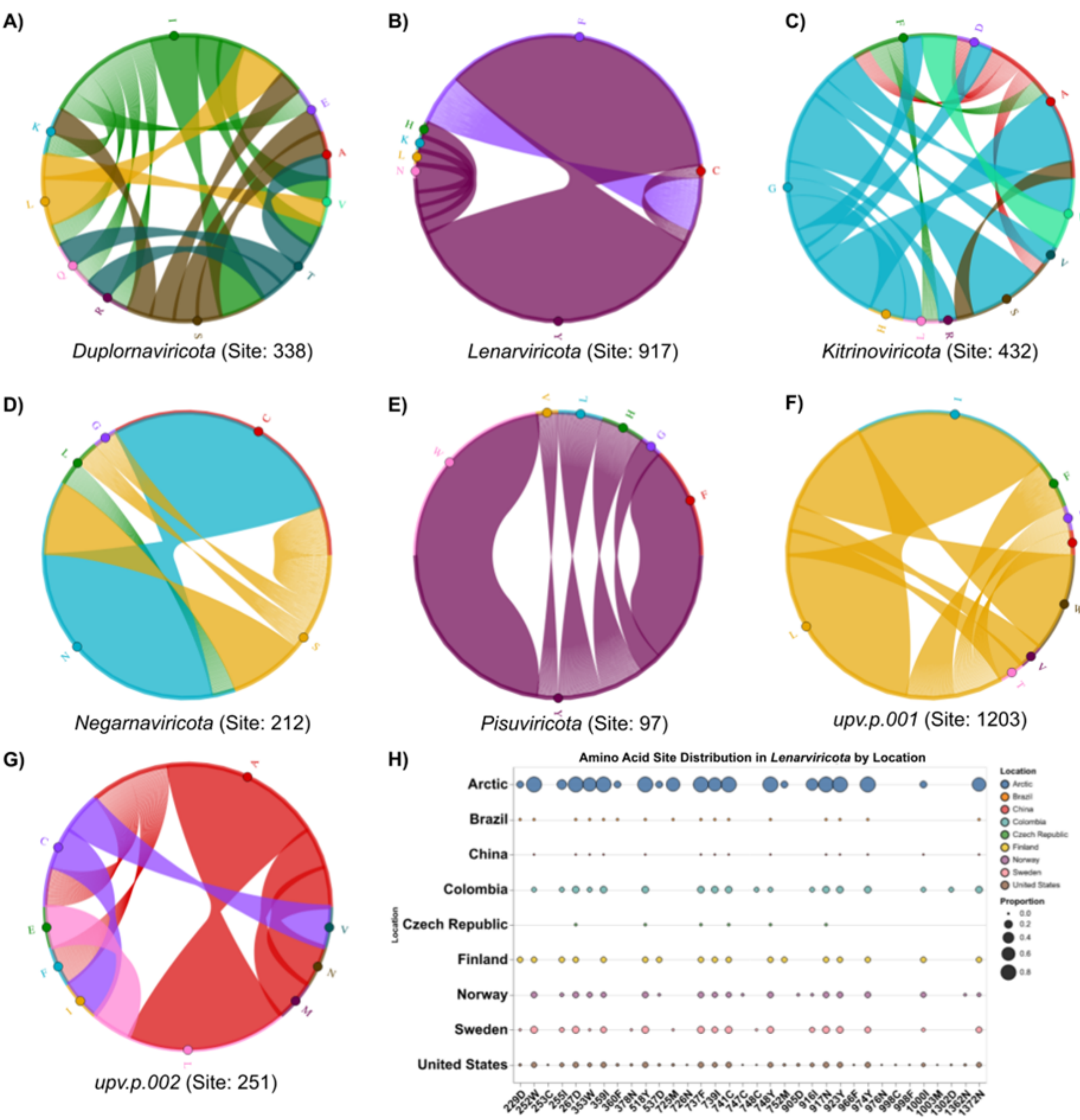


# Directional Selection Shapes Amino Acid Exchange Landscapes Across RNA Virus Phyla

Site-specific analyses reveal heterogeneous amino acid exchange patterns consistent with diverse selective pressures, with globally distributed directional selection signals exemplified in *Lenarviricota*.



# Environmental Distribution of the Urban & Peri-urban RNA Virome

*RNA virus phylum composition varies markedly across environmental sample types based on FPKM-normalized relative abundance.*

